

August 6, 2026

Dear Hiring Team,

I am writing to apply for the Junior Machine Learning Engineer position on the Global AI Team at ALS. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in Python, PyTorch, TensorFlow, Scikit-learn, data pipelines, APIs, Docker, and LLM-based automation, I am excited by the opportunity to help ALS build practical AI-enabled products and workflows for scientific and business problems.

My graduate research focused on developing scalable software and machine learning workflows for large geospatial and Earth observation datasets. I built C++ and Python systems for data processing, storage, and retrieval; automated satellite imagery preprocessing; and developed tensor-based pipelines for structured scientific data. This work required careful attention to data preparation, performance, reproducibility, documentation, and validation, all of which align closely with ALS's need for maintainable AI and ML solutions.

I also bring deep learning experience that maps directly to the role's model-development responsibilities. In my DEM super-resolution project, I trained and evaluated PyTorch ResNet models, built data collection and preprocessing workflows using Google Earth Engine and Rasterio, and assessed model quality with metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. As a Deep Learning Teaching Assistant with Neuromatch Academy, I also supported students through PyTorch tutorials covering CNNs, transformers, diffusion models, and reinforcement learning, which strengthened both my technical foundation and my ability to explain complex ML concepts clearly.

Beyond traditional ML, I have been building an agentic job search automation platform using LLM workflows, tool orchestration, Docker, n8n, OpenClaw, Google Sheets, Google Drive, and Telegram API integrations. That project has given me practical exposure to AI assistants, workflow automation, API integration, and the engineering discipline needed to connect models or model-like systems to useful products. I am especially interested in growing further in cloud AI services, MLOps, embeddings, RAG, and responsible AI within a real product team.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in machine learning, Python-based engineering, applied AI workflows, and scientific software development can contribute to ALS's Global AI Team.

Sincerely,

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